

# Causal inference for empirical dynamical systems based on persistent homology

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## Abstract

Given two correlated systems, detecting causality between them from observed data is an important but challenging task. Combining two mathematical techniques, delay coordinate embedding and persistent homology, we propose a novel causal inference method for data comprising a pair of scalar time series that are observed from two possibly coupled deterministic dynamical systems. The idea is to encode the topology of the dynamics in the form of the persistent homology of the reconstructed attractors and compare the involved systems by a metric defined on the persistent homology.

**Keywords** causal inference, empirical dynamical system, persistent homology, topological data analysis

**Research Activity Group** Topological Data Analysis

## 1. Introduction

Data analysis seeks to find hidden relationships between data, including causal relationships, in order to identify what causes the phenomenon of interest. For example, a functional brain network constructed by detecting inter-regional connections can elucidate how brain architecture supports neurophysiological dynamics [1]. Causal inference is also indispensable in order to assess the effects of interventions intended to solve various social and medical problems [2, 3].

Which causal inference method should be used depends on the nature of the system under study. The system can be static or time-dependent, and in the latter case, the system can be deterministic or probabilistic. Sometimes a reliable model exists, and other times not. When the target system is static and has a reliable model, a major approach to causal inference is structural equation modeling (SEM) [4]. When the target system involves stochastic evolution with a known model, Granger-type causal inference [5] is widely applied. The fundamental assumption behind this method is that inclusion of the observation from the causing system improves the prediction accuracy of an observation of the caused system.

In the present paper, we are interested in deterministic empirical dynamical systems with no known model, a topic which has been studied under the framework of *empirical dynamic modeling* (EDM) [6]. The underlying principle of EDM is that the behavior of the causing system should be accurately predicted by the history of the caused system in the long run. This may sound

contrary to the assumption of the methods for stochastic systems mentioned above, but the principle is theoretically guaranteed with deterministic systems by the Takens embedding theorem, which roughly states that an attractor for the true system can be reconstructed from observed time-series data. Most EDM techniques rely on the one-to-one correspondence of the true and the reconstructed attractors (see [7] for a survey). In the present paper, we propose a novel model-free causal inference method, *homological causality inference* (HCI), based on the comparison of topological invariants of the reconstructed attractors. The proposed method looks at the global topological features of attractors and does not rely on local correspondence of the attractors. More precisely, we reconstruct the attractors of the causing, caused, and total systems, and compute their *persistent homology* to encode their topological features into the persistence diagrams, which are compared using a metric defined on the persistent diagrams. The present study gives a new perspective on causal inference by focusing on the global topology of the underlying systems that are captured by homology.

## 2. Homological causality inference

Consider a system  $(A, B)$  consisting of two possibly coupled deterministic, discrete-time dynamical subsystems  $A$  and  $B$  on manifolds  $X$  and  $Y$ , respectively. When the subsystems  $A$  and  $B$  are described by maps  $A : X \rightarrow X$  and  $B : Y \rightarrow Y$ , we say that there exists no causal relation. When  $A : X \rightarrow X$  and  $B : X \times Y \rightarrow Y$ , and  $B(x_1, y) \neq B(x_2, y)$  for some  $x_1, x_2, y$ , the subsys-

tem  $A$  is said to be a cause of  $B$ . In the latter case, the reconstructed attractor for the observed time series from the caused (downstream) system  $B$  is generically homeomorphic to that for the total system  $(A, B) : X \times Y \rightarrow X \times Y$  defined by  $(A, B)(x, y) = (A(x), B(x, y))$ . This can be used to infer the causality by comparing the topologies of the reconstructed attractors. In order to implement this idea as a practical procedure, we need two ingredients. First, using the delay coordinate embedding, we reconstruct the attractors for the total system  $(A, B)$  and its subsystems  $A$  and  $B$  using time-series data  $\{\psi_A(x_t) \in \mathbb{R} \mid t \in \mathbb{N}\}$  and  $\{\psi_B(y_t) \in \mathbb{R} \mid t \in \mathbb{N}\}$  obtained through *observation maps*  $\psi_A : X \rightarrow \mathbb{R}$  and  $\psi_B : Y \rightarrow \mathbb{R}$ , respectively. Second, we compare the topologies of the reconstructed attractors by persistent homology. In what follows, we give a detailed description of the proposed method.

For a constant  $\tau > 0$  called the lag and  $E \in \mathbb{N}$  called the embedding dimension, the delay coordinate embedding of a scalar time series  $\{u_t \in \mathbb{R} \mid t \in \mathbb{N}\}$  is defined to be

$$\mathbf{M}_{u, \tau, E}(t) = (u_t, u_{t+\tau}, \dots, u_{t+(E-1)\tau}) \in \mathbb{R}^E.$$

When  $E$  is large, Takens's theorem [8] states that for a generic observation map  $\psi : X \rightarrow \mathbb{R}$  of a generic dynamical system  $A : X \rightarrow X$ , the delay coordinate embedding of  $\{\psi(x_t) \in \mathbb{R} \mid t \in \mathbb{N}\}$  is injective and the point cloud  $X_A = \mathbf{M}_{\psi, \tau, E}(\mathbb{N}) \subset \mathbb{R}^E$  approximates the attractor of the dynamical system. In practice, the optimal lag  $\tau$  is determined, for example, by looking at the average mutual information [9]. The embedding dimension  $E$  is determined, for example, by the false nearest neighbor method [10]. In what follows, we fix choices of  $\tau$  and  $E$ .

When the system consists of two subsystems  $A$  and  $B$ , and  $A$  is the cause of  $B$ , a theorem of Stark [11] states that the reconstructed attractor  $X_B$  from the delayed coordinate embedding  $\mathbf{M}_{\psi_B, \tau, E}$  of  $\{\psi_B(y_t) \in \mathbb{R} \mid t \in \mathbb{N}\}$  approximates the attractor for the total system  $(A, B)$ . Note that by Takens's theorem, the reconstructed attractor  $\mathbf{M}_{\psi_A, \tau, E}$  for the upstream system approximates the attractor for  $A$ . Furthermore, Cao [12] constructs the delayed coordinate embedding of multivariate time-series data  $\{(u_t, v_t) \in \mathbb{R}^2 \mid t \in \mathbb{N}\}$  defined by

$$\mathbf{M}_{u, v, \tau, E}(t) = (u_t, u_{t+\tau}, \dots, u_{t+(E/2-1)\tau}, \\ v_t, v_{t+\tau}, \dots, v_{t+(E/2-1)\tau}),$$

where we assume that  $E$  is an even number. We apply this construction to the observations  $\{\psi_A(x_t) \in \mathbb{R} \mid t \in \mathbb{N}\}$  and  $\{\psi_B(y_t) \in \mathbb{R} \mid t \in \mathbb{N}\}$  to obtain the reconstructed attractor  $X_{AB} = \mathbf{M}_{\psi_A, \psi_B, \tau, E}(\mathbb{N})$ . Note that  $X_{AB}$  approximates the attractor for the total system  $(A, B)$  regardless of the existence of a causal relation between  $A$  and  $B$ .

In order to compare the topologies of these reconstructed attractors, we use persistent homology [13], which extracts topological features from a point cloud  $S \subset \mathbb{R}^E$  by encoding the change of the topology of  $\bigcup_{p \in S} B_p(r)$  with regard to  $r > 0$ , where  $B_p(r)$  denotes a ball with center  $p$  and radius  $r$ . For the reconstructed attractor  $X$ , the corresponding alpha com-

plex [14]  $\mathcal{A}(X, r)$ , which is homotopy equivalent to the union of balls  $\bigcup_{p \in X} B_p(r)$ , is constructed. An increasing sequence of radius  $r$  gives rise to an increasing sequence of alpha complexes

$$\emptyset = \mathcal{A}(X, r_0) \subseteq \mathcal{A}(X, r_1) \subseteq \dots \subseteq \mathcal{A}(X, r_l).$$

We assume that  $r_l$  is sufficiently large so that  $\mathcal{A}(X, r_l)$  is homotopy equivalent to  $\mathbb{R}^E$ . Let  $\mathbb{F}_2$  be the field of two elements, and let  $H_k$  be the  $k$ -th homology functor with the coefficients in  $\mathbb{F}_2$ . Applying  $H_k$  to the above sequence, we obtain a sequence of  $\mathbb{F}_2$ -vector spaces

$$H_k(\mathcal{A}(X, r_0)) \rightarrow H_k(\mathcal{A}(X, r_1)) \rightarrow \dots \rightarrow H_k(\mathcal{A}(X, r_l)).$$

By Gabriel's theorem [15], the above sequence is decomposed uniquely as the direct sum of interval modules. The persistent diagram  $\text{Dgm}_k(X)$  is defined as the multiset of points  $(r_i, r_{j+1})$  in the plane appearing in the above direct sum decomposition.

We compare the topologies of two systems quantitatively by measuring the bottleneck distance [16] between the persistent diagrams of the corresponding alpha complexes. Before computing the bottleneck distance, we normalize the persistent diagrams by scaling to make their maximum lifetime to be one. This normalization is equivalent to scaling the point clouds and accounts for the difference in the range of the observed variables. Due to the normalization, the bottleneck distance takes values in the interval  $[0, 0.5]$ . To summarize, we define the  $k$ -th *persistence distance* for two subsystems  $A$  and  $B$  as the normalized bottleneck distance between  $\text{Dgm}_k(X_A)$  and  $\text{Dgm}_k(X_B)$ , where  $X_A$  and  $X_B$  are the reconstructed attractors from the delay coordinate embedding. The persistence distance between subsystem  $A$  and total system  $(A, B)$  is defined similarly using the reconstructed attractors  $X_A$  and  $X_{AB}$ .

Recall that when  $A$  is the cause of  $B$ , the reconstructed attractors of  $B$  and  $(A, B)$  are homeomorphic. This leads to the following definition: for a user-defined threshold  $\delta \in [0, 0.5]$ , we can say that there is a  $(\delta, k)$ -persistence causality from  $A$  to  $B$  when the  $k$ -th persistence distance of  $B$  and  $(A, B)$  is less than  $\delta$ . We can say that there is a bidirectional  $(\delta, k)$ -persistence causality between  $A$  and  $B$  when there is  $(\delta, k)$ -persistence causality in both directions. See Fig. 1 for a schematic description of the proposed method.

### 3. Numerical experiments

We apply the proposed method to synthetic time-series data generated from the spring-mass model with various coupling strengths  $(\alpha, \beta)$ . The spring-mass model is defined by

$$\frac{d^2 x(t)}{dt^2} = -x(t) + 0.7\alpha y(t), \\ \frac{d^2 y(t)}{dt^2} = -0.7y(t) + \beta x(t),$$

where the constant  $\alpha$  determines the strength of the causal influence of  $y$  on  $x$ , and  $\beta$  does the same, but in the opposite causal direction. The time series is generated by the Runge–Kutta method RK4 with initial val-

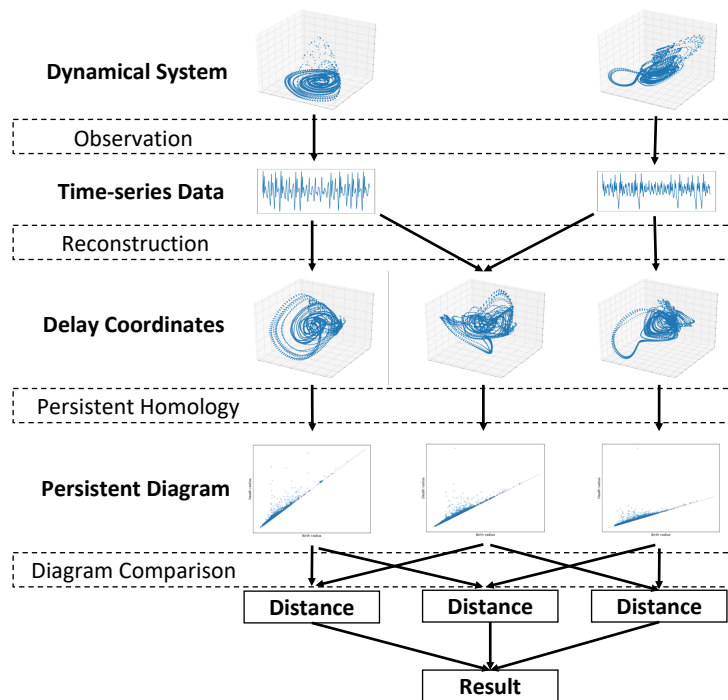


Fig. 1. Schematic diagram of homological causality inference. The pipeline of the proposed method is depicted.

ues  $(x, y, x', y') = (1, 0, 0, 1)$  for 10,000 time steps with a step size of  $\Delta t = 0.01$ . One of every 10 samples is used to form a set of 1,000 samples for our experiments. The reconstructed attractors  $X_A$  and  $X_B$  are obtained using  $x$  and  $y$  as the observations. The optimal lag  $\tau = 1$  and the optimal embedding dimension  $E = 4$  are empirically determined. The first persistent homology corresponding to the alpha complex for each attractor and the bottleneck distance for each pair of the persistent diagrams were computed by the program GUDHI [17, 18]. The first persistent homology is chosen because higher persistent homology was found to be almost trivial in the current experiment.

Table 1 shows the pairwise first persistence distances between  $X_A$ ,  $X_B$ , and  $X_{AB}$  for coupling strengths spanning  $0.0 \leq \alpha \leq 0.8$  and  $0.0 \leq \beta \leq 0.8$ . The entries for which  $(\delta, 1)$ -persistence causality with  $\delta = 0.4$  agree with the actual causal relation are highlighted. Here, we regard  $\alpha > 0$  (respectively,  $\beta > 0$ ) as the existence of causal influence from  $B$  to  $A$  (respectively, from  $A$  to  $B$ ).

In the cases of bidirectional causality ( $0.1 \leq \alpha \leq 0.8$ ,  $0.1 \leq \beta \leq 0.8$ ), the  $(\delta, 1)$ -persistence causality agrees with the true causality with an accuracy of 64.1%. In the cases with strong coupling ( $\alpha \geq 0.3$ ,  $\beta > 0.21 = 0.7 \cdot 0.3$ ), the accuracy increases to 82.9%, which indicates that  $(\delta, 1)$ -persistence causality is better at detecting stronger causal relations. In the cases of unidirectional causality from  $B$  to  $A$  ( $0.1 \leq \alpha \leq 0.8$ ,  $\beta = 0$ ), the accuracy is 87.5%, whereas the accuracy for the opposite unidirectional causality is 75.0%. Moreover, the  $(\delta, 1)$ -persistence causality agrees with the true causality when  $A$  and  $B$  are independent, that is, when  $\alpha = \beta = 0$ .

## 4. Discussion

In the present study, we propose a causal inference method for time-series data based on the topology of reconstructed attractors from the delay coordinate embedding. Existing methods based on delay coordinate embedding rely on that fact that the time index gives a one-to-one (or local) correspondence between points on the reconstructed attractors. Instead of this local correspondence, we look only at the global topological features of the reconstructed attractors that are captured by the persistent homology. Note that persistent homology is isometry-invariant, but not homeomorphism-invariant, and the bottleneck distance varies even for homeomorphic attractors. Nonetheless, we observed in our experiment that the bottleneck distance was a robust measurement for the topological similarity of reconstructed attractors.

The concept of our novel approach is validated by numerical experiments with a simple system. However, a practical implementation and its applicability to real-world dynamical systems should be examined in a future study. For example, search criteria for the optimal lag, embedding dimension, and threshold for persistence distance should be established. Furthermore, computational complexity can be prohibitive when the embedding dimension is large. This is because the construction of the alpha complex at high dimensions is computationally intractable [19]. In such a case, we have to use alternative complexes such as the Vietoris–Rips complex, which is not homotopy equivalent to the alpha complex.

Finally, we emphasize that the proposed method is not necessarily superior to existing EDM techniques, because the proposed method does not use local information, which may include important information on the time-local behavior of the underlying system. How-

Table 1.  $(\delta, 1)$ -persistence distance with  $\delta = 0.4$ . Pairwise first persistence distances between  $X_A$ ,  $X_B$ , and  $X_{AB}$  are shown. Entries for which the  $\delta$ -persistence causality with  $\delta = 0.4$  agrees with the actual causal relation are highlighted.

| $\alpha \backslash \beta$ |      | .0         | .1         | .2         | .3         | .4         | .5         | .6         | .7         | .8         |
|---------------------------|------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| .0                        | A-B  | <b>.50</b> | .50        | <b>.50</b> | <b>.50</b> | <b>.50</b> | .50        | <b>.50</b> | <b>.50</b> | <b>.50</b> |
|                           | B-AB | <b>.50</b> | .44        | <b>.20</b> | <b>.21</b> | <b>.29</b> | .43        | <b>.35</b> | <b>.36</b> | <b>.37</b> |
|                           | A-AB | <b>.50</b> | .50        | <b>.50</b> | <b>.50</b> | <b>.50</b> | .50        | <b>.50</b> | <b>.50</b> | <b>.50</b> |
| .1                        | A-B  | .50        | <b>.15</b> | <b>.30</b> | .42        | .25        | <b>.34</b> | .43        | .24        | .47        |
|                           | B-AB | .50        | <b>.38</b> | <b>.19</b> | .37        | .49        | <b>.31</b> | .23        | .42        | .32        |
|                           | A-AB | .41        | <b>.38</b> | <b>.38</b> | .49        | .49        | <b>.29</b> | .47        | .45        | .49        |
| .2                        | A-B  | <b>.50</b> | .46        | .48        | .43        | <b>.28</b> | .00        | .37        | .42        | <b>.22</b> |
|                           | B-AB | <b>.50</b> | .29        | .34        | .21        | <b>.29</b> | .50        | .23        | .28        | <b>.13</b> |
|                           | A-AB | <b>.25</b> | .32        | .42        | .42        | <b>.21</b> | .50        | .42        | .49        | <b>.31</b> |
| .3                        | A-B  | <b>.49</b> | <b>.39</b> | <b>.25</b> | <b>.30</b> | .40        | <b>.31</b> | <b>.14</b> | <b>.29</b> | <b>.20</b> |
|                           | B-AB | <b>.50</b> | <b>.34</b> | <b>.32</b> | <b>.31</b> | .26        | <b>.17</b> | <b>.28</b> | <b>.32</b> | <b>.24</b> |
|                           | A-AB | <b>.36</b> | <b>.27</b> | <b>.35</b> | <b>.27</b> | .46        | <b>.20</b> | <b>.27</b> | <b>.34</b> | <b>.24</b> |
| .4                        | A-B  | <b>.46</b> | .40        | .41        | <b>.29</b> | <b>.21</b> | <b>.33</b> | <b>.25</b> | <b>.11</b> | .11        |
|                           | B-AB | <b>.49</b> | .40        | .10        | <b>.19</b> | <b>.13</b> | <b>.28</b> | <b>.28</b> | <b>.32</b> | .48        |
|                           | A-AB | <b>.29</b> | .23        | .35        | <b>.30</b> | <b>.23</b> | <b>.10</b> | <b>.21</b> | <b>.32</b> | .49        |
| .5                        | A-B  | <b>.44</b> | <b>.15</b> | .42        | <b>.18</b> | .40        | <b>.17</b> | <b>.20</b> | <b>.13</b> | <b>.11</b> |
|                           | B-AB | <b>.50</b> | <b>.18</b> | .29        | <b>.23</b> | .47        | <b>.34</b> | <b>.27</b> | <b>.29</b> | <b>.22</b> |
|                           | A-AB | <b>.19</b> | <b>.17</b> | .48        | <b>.15</b> | .26        | <b>.33</b> | <b>.29</b> | <b>.16</b> | <b>.26</b> |
| .6                        | A-B  | <b>.50</b> | .46        | <b>.12</b> | <b>.08</b> | <b>.37</b> | <b>.13</b> | .29        | .49        | <b>.10</b> |
|                           | B-AB | <b>.49</b> | .46        | <b>.22</b> | <b>.29</b> | <b>.11</b> | <b>.35</b> | .36        | .29        | <b>.26</b> |
|                           | A-AB | <b>.26</b> | .30        | <b>.34</b> | <b>.31</b> | <b>.39</b> | <b>.37</b> | .42        | .41        | <b>.30</b> |
| .7                        | A-B  | <b>.50</b> | .00        | <b>.20</b> | <b>.26</b> | <b>.15</b> | <b>.18</b> | <b>.29</b> | <b>.22</b> | <b>.21</b> |
|                           | B-AB | <b>.46</b> | .47        | <b>.31</b> | <b>.29</b> | <b>.27</b> | <b>.33</b> | <b>.16</b> | <b>.29</b> | <b>.29</b> |
|                           | A-AB | <b>.28</b> | .47        | <b>.35</b> | <b>.32</b> | <b>.31</b> | <b>.30</b> | <b>.26</b> | <b>.33</b> | <b>.27</b> |
| .8                        | A-B  | <b>.50</b> | .48        | <b>.14</b> | <b>.14</b> | <b>.16</b> | <b>.36</b> | <b>.17</b> | <b>.29</b> | .46        |
|                           | B-AB | <b>.45</b> | .50        | <b>.17</b> | <b>.28</b> | <b>.26</b> | <b>.15</b> | <b>.15</b> | <b>.17</b> | .29        |
|                           | A-AB | <b>.29</b> | .30        | <b>.20</b> | <b>.30</b> | <b>.29</b> | <b>.40</b> | <b>.17</b> | <b>.23</b> | .39        |

ever, our idea of comparing the topologies of three reconstructed attractors provides a new approach in the study of causal inference.

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